

Futureproof

Equity Preservation Mortgage v14b

Wholesale Funding

Credit Risk Analysis & Portfolio Economics | March 2025

March 2025

CONFIDENTIAL — For Discussion Purposes Only

Executive Summary

FutureProof seeks a wholesale funding partnership to originate Equity Preservation Mortgages — a new class of residential mortgage product with unique risk characteristics. The v14b quantitative model demonstrates that EPMs, structured with P&I; repayment and index-linked ETF allocation, achieve a portfolio-level Probability of Claim (PoC) of just 0.55% at year 30 after the Payments Waterfall is applied. The product incorporates stochastic interest rates, explicit hedging costs, and portfolio-level diversification with cross-subsidisation.

Key Points for Wholesale Funders

- **Extremely low claim risk** — Portfolio PoC of 0.55% at year 30 after Payments Waterfall (ETF selldown, cross-subsidisation from surplus loans). Individual PoD of 14.2% is a balance sheet snapshot, not the claim probability.
- **Fully collateralised** — Secured against residential property (\$2M+ values) with 60-80% LVR
- **Self-liquidating** — P&I; structure amortises loan to zero over tenure; no balloon risk
- **No early exit risk** — Unique run-off mechanism: no voluntary prepayments or early terminations until all loan costs paid from the investment account
- **Investment portfolio as additional collateral** — Index-linked ETF provides secondary security above the property
- **Surplus split 50/50** — Funder receives 50% of surplus at maturity, creating alignment
- **Insurance paid upfront** — LMI (90% of loss) and tail risk premiums are paid upfront at loan origination, providing full coverage from day one

Product Overview for Funders

Feature	Detail	Funder Relevance
Collateral	Residential property (\$2M+)	Prime residential, first mortgage
LVR	60-80%	Conservative LVR range
Loan Type	Principal & Interest (default)	Self-amortising, no balloon
Loan Size	\$1.0-1.6M typical	Institutional scale
Tenure	Up to 30 years	Long-duration, stable AUM
Borrower Profile	Age 55-75, high-equity homeowners	Low-risk demographic
Investment	S&P500; index-linked ETFs (BlackRock/SpiderRock)	Passive index + dynamic hedging
Interest Coverage	Paid from investment returns	Not reliant on borrower income
Insurance	LMI + tail risk coverage	Additional loss protection

How the Funder Gets Paid

- Wholesale margin (2.0%) applied above the cash rate — paid quarterly from the equity investment portfolio
- During market downturns, interest may be temporarily deferred (capitalised) — NOT defaulted. In any given year, the median mortgage requires zero holidays; deferred interest is repaid as markets recover
- P&I; structure means loan balance is continuously reducing — declining credit exposure over time
- At exit: loan repaid from investment portfolio and/or property sale
- Insurance covers any deficit between investment portfolio and loan balance

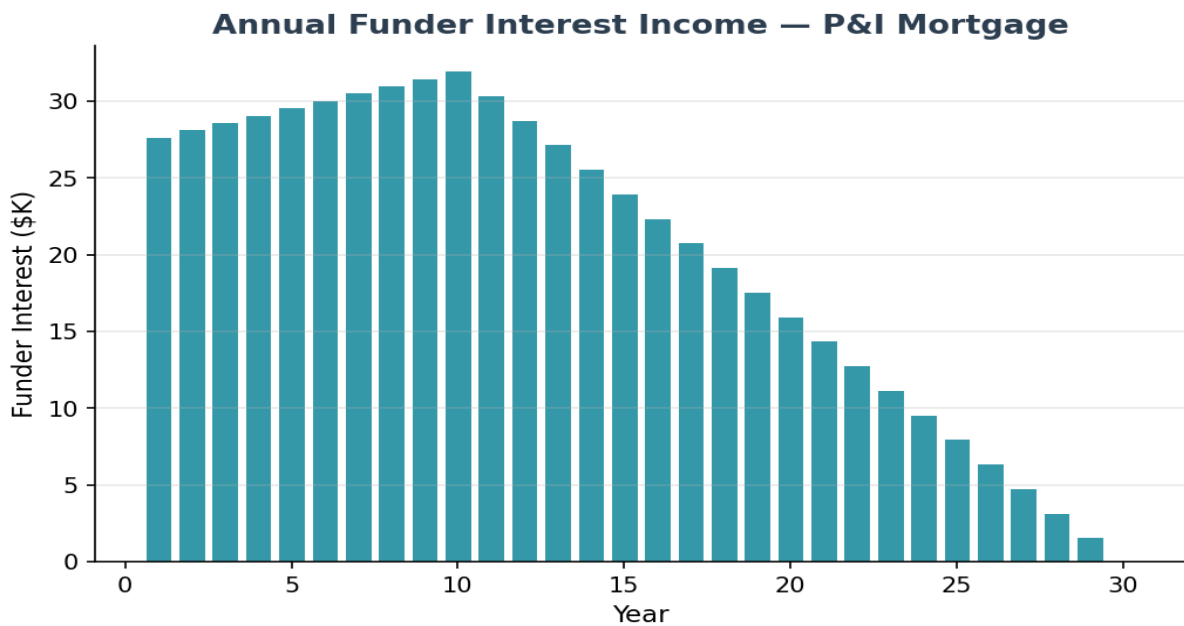
Credit Risk Analysis

The v14b model provides detailed credit risk metrics. The critical distinction is between PoD (balance sheet view) and PoC (actual insurance claim probability). PoC is the metric relevant to funders and reinsurers:

Credit Metric	Value	Context
Portfolio PoC (Yr 30)	0.55%	After Payments Waterfall — the key metric
Individual PoD (Yr 30)	14.2%	Balance sheet snapshot (not claim probability)
Cond. Expected Deficit	\$592,400	Average deficit when deficit occurs (before waterfall)
Insurance Premium (PV)	\$28,798	Based on discounted expected deficit (S3) at 4.4%
Insurance + Loading	\$32,929	Paid upfront at loan origination
Mean Surplus at Maturity	\$1,690,289	Split 50/50 FutureProof/Funder
Property Security	\$2M+ (first mortgage)	Covers loan + deficit in all scenarios

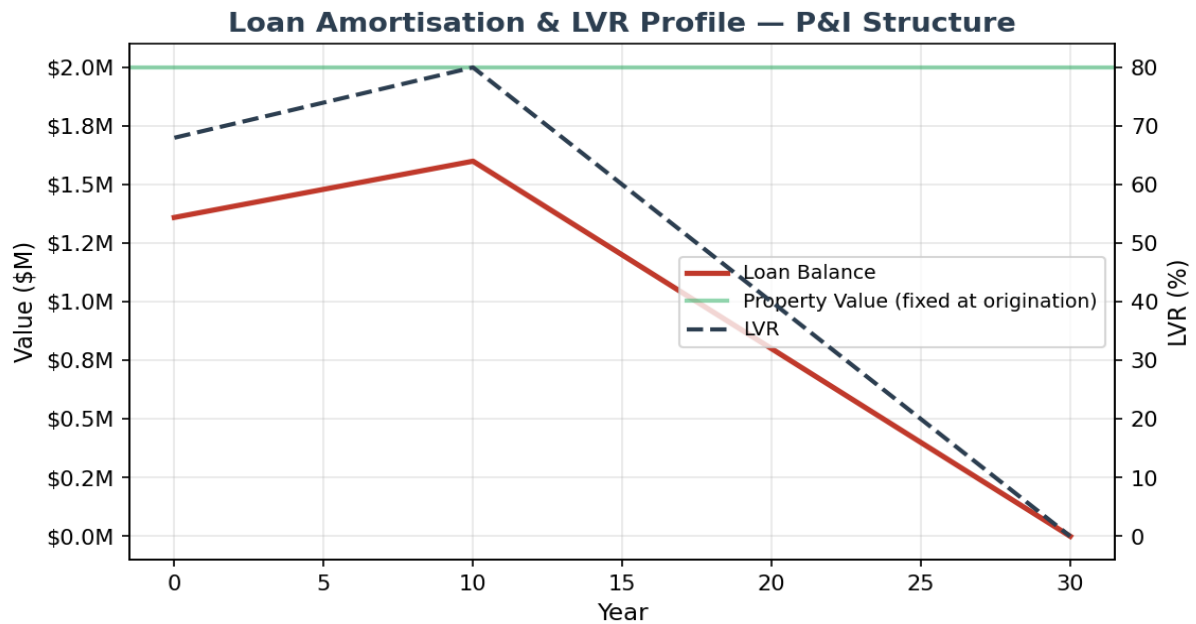
The portfolio PoC of 0.55% at year 30 is comparable to or better than prime residential mortgage default rates (typically 0.5-1.5% over life). This low PoC is achieved through the Payments Waterfall: before any insurance claim on an expiring loan, ETFs are sold down, then surpluses from other open mortgages cross-subsidise any deficit. Only the residual net deficit (if any) triggers a claim. Additionally, the funder receives 50% of surplus at maturity, and the unique run-off mechanism prevents early exits that could truncate compounding.

Funder Cash Flow Profile



The chart shows mean annual funder payments (interest margin + principal) over the EPM lifecycle. Years 1-10 show cash flow in the \$40-70K range as the annuity payments reduce available investment income. After year 10 (when annuity ceases), funder payments stabilise at \$100-113K per year. The year 30 value is negative because it represents the final loan balance repayment at maturity. Note: there is no early exit risk — the run-off mechanism prevents voluntary prepayments until all loan costs are paid, giving the funder predictable, long-duration cash flows.

Loan Amortisation & LVR Profile



The chart illustrates the declining funder exposure under the P&I structure. The loan balance amortises to zero over 30 years against the fixed property value at origination. House prices are deterministic in this model — they set the LVR at origination but do not affect ongoing mechanics (this is not a shared appreciation mortgage). The LVR declines purely through P&I amortisation, meaning the wholesale funder's security position improves every single year regardless of property market movements.

Year	Loan Balance	Est. Property Value	LVR
0	\$1,360,000	\$2,000,000	68.0%
5	\$1,263,240	\$2,318,548	54.5%
10	\$1,143,474	\$2,687,833	42.5%
15	\$978,147	\$3,115,935	31.4%
20	\$749,927	\$3,612,222	20.8%
25	\$434,887	\$4,187,556	10.4%
30	\$0	\$4,854,525	0.0%

Collateral & Security Structure

First Mortgage Security

- Registered first mortgage over residential property
- Minimum property value: \$1.5M (typically \$2M+)
- Maximum LVR at origination: 80% (typical 68.0%)
- Property independently valued at origination

Investment Portfolio as Additional Security

- Loan proceeds held in mortgage offset account, invested by BlackRock in S&P500 index-linked ETFs
- Investment portfolio provides secondary collateral (mean value \$2.7M at year 15)
- In the P&I; structure, the loan amortises while the investment compounds
- Index-linked ETF mechanics (cap/floor) provide defined outcome protection

Insurance Structure

- Insurance premiums (LMI + tail risk) are paid upfront at loan origination — providing full coverage from day one
- LMI covers 90% of loss; worst 10% quantile transferred as tail risk to reinsurance
- Premium calculated on discounted expected deficit (S3), discounted at 4.4% (cash rate mean)
- No insurance claim can be made until a loan expires — PoD before expiry is irrelevant
- Portfolio PoC at year 30 is 0.55% after Payments Waterfall — well below typical reinsurance thresholds

Payments Waterfall at Loan Expiry

1. Sell down ETFs from the expiring loan's investment account
2. Cross-subsidise from surpluses of other open mortgages in the portfolio
3. Only then is the net deficit known — this is the actual Probability of Claim (PoC)
4. If net deficit remains: LMI covers 90%, reinsurance covers tail risk (worst 10% quantile)
5. Loan balance repaid to wholesale funder (priority claim)
6. Surplus split 50/50 between FutureProof and Mortgage Funder

Portfolio Diversification Benefits

Why a Book of EPMs is Safer Than a Single EPM

- The Payments Waterfall enables cross-subsidisation: surpluses from performing loans offset deficits in expiring loans
- This is why portfolio PoC (0.55%) is dramatically lower than individual PoD (14.2%)
- Different origination dates mean different equity market entry points — vintage diversification
- P&I; amortisation means older EPMs have very low residual exposure
- v14b models portfolio of 10,000 EPMs across 4 tenure bands (15, 20, 25, 30 years)
- Multi-vintage structure means the portfolio always contains a mix of seasoned (low-risk) and new EPMs
- Run-off mechanism prevents early exits — every loan contributes to portfolio cross-subsidisation for its full term

Vintage Diversification — Why Not All Mortgages Are Affected Equally

A critical feature of the EPM portfolio is that not all mortgages are affected equally in a market downturn. Over a 30-year period, EPMs are originated at different points in time — each vintage enters the equity market at a different level. A downturn that causes a deficit for mortgages originated at market peaks may have little or no impact on mortgages originated years earlier (which have already accumulated substantial surplus) or years later (which enter at lower market levels with more room to grow). This vintage diversification means that even in severe downturns, the majority of the portfolio remains in surplus, providing a deep pool for cross-subsidisation of the small number of affected mortgages approaching expiry.

Good vs Bad Pathways — The Mathematics of Cross-Subsidisation

The stochastic analysis of S&P500; returns over 30 years reveals a fundamental asymmetry that makes portfolio cross-subsidisation highly effective. Of the 50,000 simulated pathways, approximately **86% deliver sufficient or excess returns** to cover all loan costs (surplus at maturity), while only **~14% result in a deficit** (individual PoD of 14.2%). At the portfolio level, after the Payments Waterfall applies cross-subsidisation from surplus loans, the net claim probability drops to just **0.55% (PoC)**. This means that for every loan in deficit, there are roughly 6 loans in surplus available to cross-subsidise — making the Payments Waterfall an extraordinarily effective risk mitigation mechanism before any reinsurance claim is required.

Illustrative Portfolio Build-Up

Year	New EPMs	Total Active	Total AUM	Avg LVR
1	100	100	\$136M	68.0%
3	300	700	\$945M	58%

Year	New EPMS	Total Active	Total AUM	Avg LVR
5	500	2,000	\$2.7B	50%
10	500	4,000	\$5.4B	42%

The Partnership Proposition

What FutureProof Offers the Wholesale Funder

- Access to a new, growing asset class with 0.55% portfolio PoC — comparable to prime residential
- Long-duration, stable AUM — run-off mechanism ensures no early exits
- Self-amortising P&I; loans — credit quality improves every year
- First mortgage security over prime residential property (\$2M+)
- 50% of surplus at maturity — aligning funder and originator incentives
- Institutional fund management (BlackRock) removes counterparty risk on investments
- Payments Waterfall and insurance layers provide multi-level loss protection

What FutureProof Seeks

- Competitive wholesale margin (target: 1.5-2.0% above cash rate)
- Commitment to fund an initial pilot of 50-100 EPMs (\$67-135M)
- Scalable facility to support growth to \$2.5B+ over 5 years
- Collaborative approach to product development and risk sharing

The Win-Win

- Lower margin -> viable IO tier -> larger addressable market -> more AUM for the funder
- At 2.0% margin on \$2.5B portfolio = \$50M annual wholesale revenue
- At 3.0% on a smaller \$1.0B portfolio = \$30M — lower total revenue despite higher per-unit margin
- Long-duration, low-risk, self-amortising residential assets are exactly what bank balance sheets want
- ESG-positive product supports responsible lending and retirement security narratives

Methodology

Component	Detail
Simulation engine	50,000-path Monte Carlo, annual time steps, 30-year tenure
Risk factors	Stochastic equity returns (lognormal) + stochastic cash rates (OU)
Equity-rate correlation	0.21 (appropriate for 30-year horizon; long-term marginally positive)
Index-linked ETF	Cap/floor on quarterly returns (+20%/-20%)
House prices	Deterministic (sets LVR at origination only; not a shared appreciation mortgage)
Prepayment/Early exit	Run-off mechanism: no exit until all loan costs paid
Insurance pricing	Based on discounted expected deficit (S3); discounted at 4.4% (cash rate mean)
Insurance structure	LMI covers 90% of loss; worst 10% quantile = tail risk (reinsurance)
Payments Waterfall	ETF selldown → cross-subsidise from surpluses → net deficit → PoC
Surplus split	50/50 FutureProof and Mortgage Funder
Portfolio model	10,000 EPMs, 4 tenure bands, 6 vintage years (2026-2031)